

Statistical Arbitrage

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October 4, 2026

Problem 1: The OU Process

1. **Solving the SDE** We first expand the difference $d(e^{\theta t} X_t)$ using the product rule. This is equivalent to the method of integrating factor for ODEs.

$$\begin{aligned}dX_t &= \theta(\mu - X_t)dt + \sigma dW_t, \quad \theta, \sigma > 0 \\d(e^{\theta t} X_t) &= e^{\theta t} dX_t + \theta e^{\theta t} X_t dt \\d(e^{\theta t} X_t) &= e^{\theta t} (\theta(\mu - X_t)dt + \sigma dW_t) + \theta e^{\theta t} X_t dt\end{aligned}$$

We apply Itô's lemma and integrate the SDE considering that $dW \sim N(0, dt)$.

$$\begin{aligned}d(e^{\theta t} X_t) &= \theta e^{\theta t} \mu dt + \sigma e^{\theta t} dW_t \\e^{\theta t} X_t - X_0 &= \mu(e^{\theta t} - 1) + \sigma \int_0^t e^{\theta s} dW_s \\X_t &= \mu - (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s\end{aligned}$$

This is the solution to our original SDE.

2. **Demonstrate X is gaussian** We begin by dividing our solution X_t into deterministic and stochastic components, α , and β respectively. We find that β is a zero-mean gaussian with a computable variance.

$$\begin{aligned}X_t &= \alpha + \beta, \quad \alpha = \mu - (X_0 - \mu)e^{-\theta t}, \quad \beta = \sigma \int_0^t e^{-\theta(t-s)} dW_s \\ \beta &\sim N(0, \sigma^2 \int_0^t e^{-2\theta(t-s)} ds) \\ &\sim N\left(0, \frac{\sigma^2}{2\theta} [1 - e^{-2\theta t}]\right)\end{aligned}$$

Since X_t is composed of a gaussian, we know that X_t is gaussian. We now solve for the mean and variance.

$$\begin{aligned}E(X_t) &= E\left(\mu - (X_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s\right) \\ &= \mu - (X_0 - \mu)e^{-\theta t} + E\left(\sigma \int_0^t e^{-\theta(t-s)} dW_s\right) \\ &= \mu - (X_0 - \mu)e^{-\theta t}\end{aligned}$$

This is exactly equal to the deterministic component, α , of our original random variable. Next we look

at the variance.

$$\begin{aligned}
\text{Var}(X_t) &= E(X_t^2) - E(X_t)^2 \\
&= E(\alpha^2 + 2\alpha\beta + \beta^2) - \alpha^2 \\
&= E(\beta^2) \\
&= \frac{\sigma^2}{2\theta} [1 - e^{-2\theta t}]
\end{aligned}$$

Using these mean and variance we find that the stationary distribution for $\lim_{t \rightarrow \infty} X_t$ is

$$\lim_{t \rightarrow \infty} X_t \sim N\left(\mu, \frac{\sigma^2}{2\theta}\right)$$

3. **Half-life of a deviation** First we define a deviation as a perturbation from an initial condition. Since this is a stochastic timeseries, we can inject this deviation at any point in the timeseries. We define its half-life λ as how long it takes for that deviation to decrease to half of its original value.

$$\begin{aligned}
X_t|X_0 + d &= \mu + (X_0 + d - \mu)e^{\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s \\
X_t|X_0 + d &= (X_t|X_0) + de^{-\theta t}
\end{aligned}$$

We find that the deviation follows an exponential decay curve given by $e^{-\theta t}$. The half life of this is easy to measure, we have

$$\begin{aligned}
\frac{1}{2} &= e^{-\theta \lambda} \\
\lambda &= \frac{\ln(2)}{\theta}
\end{aligned}$$

4. **Exact Discretization:** We find the exact discretization by choosing $t = \Delta t$ in the original solution to the SDE. We find

$$\begin{aligned}
X_{n+1}|X_n &= \mu + (X_n - \mu)e^{-\Delta t \theta} + \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s \\
&= \mu(1 - e^{-\Delta t \theta}) + X_n e^{-\Delta t \theta} + \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s \\
&= a + bX_n + \epsilon_k
\end{aligned}$$

Here we have shown that $a = \mu(1 - b)$, $b = e^{-\Delta t \theta}$ and $\epsilon_k = \int_0^{\Delta t} e^{-\theta(\Delta t-s)} dW_s$. We have already shown (using Δt to replace t) that $\epsilon_k \sim N\left(0, \frac{\sigma^2}{2\theta}(1 - b^2)\right)$. We invert these equations to find,

$$\begin{aligned}
\theta &= -\frac{\ln(b)}{\Delta t} \\
\mu &= \frac{a}{1 - b} \\
\sigma &= -\frac{2\text{Var}(\epsilon_k) \ln(b)}{\Delta t(1 - b^2)}
\end{aligned}$$

Problem 2: Simulation and Estimation

1. Simulating Euler-Maruyama

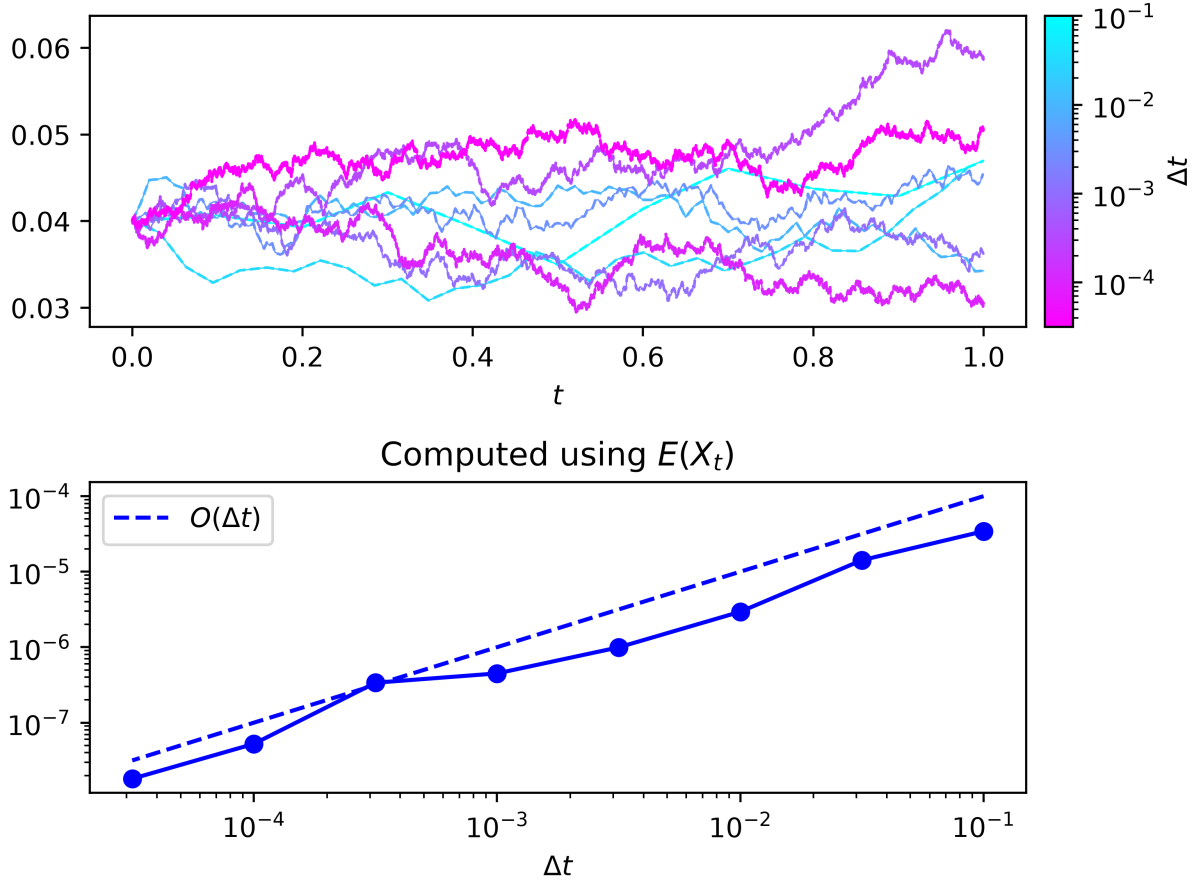


Figure 1: Several realizations of $X_t|X_0$ with varying dt

2. **Maximum-Likelihood Estimation** Derive MLE of (θ, μ, σ) . Show it is equivalent to the least squares regression of X_{k+1} on X_k .

We begin by identifying the probability/likelihood of X_{n+1} given X_n which is gaussian. In order to simplify the equation, we denote $\mu_\star = E(X_{n+1}|X_n)$ and $\sigma_\star = \text{Var}(X_{n+1}|X_n)$.

$$P_n = P(X_{n+1}|X_n; \theta, \mu, \sigma) = \frac{1}{\sigma_\star \sqrt{2\pi}} \exp\left(-\frac{(X_{n+1} - \mu_\star)^2}{2\sigma_\star^2}\right)$$

Thus we can assemble the (log) likelihood function L by taking the product of these possibilities for X_i for all $i \in [0, N_t]$. We have,

$$\begin{aligned} L(\mathbf{X}_{n+1}|\mathbf{X}_n; \theta, \mu, \sigma) &= \prod_{i=0}^{N_t-1} P_i \\ \ln(L) &= \sum_{i=0}^{N_t-1} \ln(P_i) \\ \ln(L) &= -N_t \ln(\sigma_\star \sqrt{2\pi}) - \sum_i \frac{(X_{n+1} - \mu_\star)^2}{2\sigma_\star^2} \end{aligned}$$

Without loss of generality, we will study the maximum likelihood of $X_1|X_0$ as the inclusion of additional

terms will only add rows to the eventual linear regression problem.

$$\begin{aligned}\mathcal{L} &= \ln(L(X_1|X_0; \theta, \mu, \star)) = -\ln(\sigma_\star \sqrt{2\pi}) - \frac{(X_1 - \mu_\star)^2}{2\sigma_\star^2} \\ \mathcal{L}_{\mu_\star} &= \frac{(X_1 - \mu_\star)}{\sigma_\star^2} = 0 \\ \mathcal{L}_{\sigma_\star} &= -\frac{1}{\sigma_\star} + \frac{(X_1 - \mu_\star)^2}{\sigma_\star^3} = 0\end{aligned}$$

From these two partial derivatives of the log-likelihood function we can solve for θ, μ, σ . From the partial $\mathcal{L}_{\mu_\star}$, we find

$$\begin{aligned}\frac{(X_1 - \mu_\star)}{\sigma_\star^2} &= 0 \\ X_1 - \mu - (X_0 - \mu)e^{-\Delta t \theta} &= 0 \\ X_1 &= X_0 e^{-\Delta t \theta} - \mu(1 - e^{\Delta t \theta})\end{aligned}$$

This reduces the maximum-likelihood estimation for θ, μ to a linear regression from X_0 to X_1 , i.e. we obtain (a, b) from the linear update function. The slope and y-intercept of that regression yield θ and μ .

Next we look at the equation for $\mathcal{L}_{\sigma_\star}$:

$$\begin{aligned}\frac{(X_1 - \mu_\star)^2}{\sigma_\star^3} &= \frac{1}{\sigma_\star} \\ (X_1 - X_0 e^{-\Delta t \theta} - \mu(1 - e^{\Delta t \theta}))^2 &= \sigma_\star^2\end{aligned}$$

From this equation, we find that square of the residual for this linear fit is exactly equal to the “modified” variance for this update probability. To generalize this for N_t timesteps, we turn the linear regression into a least squares regression problem where we fit a single a and b (c.f. problem 1.4) to fit each timestep update. Most modern Least-Squares solvers return the sum of the squared residuals as an output alongside the linear regression. In order to compute the “modified” variance ($\sim \text{Var}(\epsilon_k)$), we divide the the sum of squared residuals by $N_t - 1$. We now have $(a, b, \text{Var}(\epsilon_k))$, and can invert the equations to solve for (θ, μ, σ) .

3. Analyze the effectiveness of θ .

Problem 3: Pair Selection and Spread

Problem 4: Time Varying Hedge Ratio

1. **Kalman-Filter** We begin by writing the state-space model,

$$\ln(P_A) = \alpha_t + \beta_t \ln(P_B) + v_t$$

where α_t and β_t are described by random walks and v_t is a zero-mean gaussian with variance R describing some noise in the price data. We will now look at the evolution of (α_t, β_t) . We know that increments to these are i.i.d. white noise. Thus,

$$\begin{bmatrix} \alpha_t \\ \beta_t \end{bmatrix} = \begin{bmatrix} \alpha_{t-1} \\ \beta_{t-1} \end{bmatrix} + dW$$

Thus we expect noise to be injected into the state model with variance Δt . Let us assume that we have already done the task of fitting (α_t, β_t) for some $t \leq t_0$ with confidence P . We would then have the following prediction for t_1 :

$$\hat{\alpha}_t = (\hat{\alpha}_t, \hat{\beta}_t) = (\alpha_{t-1}, \beta_{t-1}) \hat{P}_t = P_{t-1} + Q$$

We then look at how closely these α_t, β_t fit the observation model: $\ln(P_A) = \mathbf{H}_t \cdot \boldsymbol{\alpha}_t$. We obtain the error in our observations given by:

$$e_t = \ln(P_A) - \mathbf{H}_t \cdot \hat{\boldsymbol{\alpha}}_t$$

This is coupled with the uncertainty/confidence of our state model being injected into our observations. We find the confidence at this level being,

$$oe_t = \mathbf{H}_t \cdot (\hat{\mathbf{P}}_t \mathbf{H}_t) + R$$

and finally the ratio between our state and observation confidence being equivalent to the kalman gain K_t .

$$\mathbf{K}_t = \frac{1}{oe_t} \hat{\mathbf{P}}_t \mathbf{H}_t$$

Using the Kalman gain we can correct our model, we have

$$\begin{aligned} \boldsymbol{\alpha}_t &= \hat{\boldsymbol{\alpha}}_t + \mathbf{K}_t e_t \\ \mathbf{P}_t &= (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t^T) \hat{\mathbf{P}}_t \end{aligned}$$

Therefore we have a nice recursion, if we know our Kalman filter $\boldsymbol{\alpha}_t$ and $\mathbf{P}_t$ we can always update the filter with another timestep following the above scheme. Doing this over, the entirety of the given timeseries allows us to find a hedge ratio β_t which is closer to the true hedge ratio at a given time t . Note that that Kalman filter can be updated as new data is presented and therefore so can our spread. As new data is introduced to our system, after updating the hedge-ratio/spread, we have a new timestep to fit the SDE. A simple recursive least-squares solution is performed to update μ, θ, σ .

2. Comparing static and Kalman hedge ratios

Problem 5: Trading Rule and Backtest

1. Trade the Standardized Set
2. Run the trading scheme on the test data
3. Diagnose and analyze a broken pair